Scalable Multi-Modal Differential Diagnosis of Neurological Diseases via Attention-Weighted Hypergraph Convolutional Networks

Jacob Mongold
Rochester Institute of Technology
jcm4822@g.rit.edu

Abraham Furlan
Rochester Institute of Technology
ajf3634@g.rit.edu

Daniel Noone
Rochester Institute of Technology
dan7492@g.rit.edu

Abstract

Differential diagnosis of neurodegenerative diseases such as Alzheimer's disease and vascular dementia is a clinically difficult problem due to overlapping symptom profiles and fundamentally different underlying pathologies. Existing graph neural network approaches to disease classification rely on pairwise patient similarity edges and fixed hyperedge heuristics that do not adapt during training, limiting their ability to model the higher-order inter-patient relationships that distinguish overlapping disease classes. We propose a Hyper-Fused Neuro-Net that combines structural MRI features as node-level representations with clinical tabular data as dynamically weighted hyperedges in a population graph, fused with a cross-attention transformer to allow imaging and clinical modalities to inform each other during classification. We evaluate our approach on the NACC Uniform Data Set across multiple neurodegenerative disease classes and benchmark against established graph-based baselines, targeting classification accuracy above 90%.

1 Introduction

Alzheimer's disease, vascular dementia, and related neurodegenerative conditions are increasingly studied in the machine learning literature. Graph neural networks, including graph convolutional networks Kipf and Welling [2017] and graph attention networks, have become the dominant approach, particularly through population graphs where each patient is a node and edges encode clinical similarity Zhou et al. [2023]. This lets models leverage inter-patient relationships rather than treating each subject in isolation.

However, current architectures have two meaningful limitations. First, most models assign edge weights uniformly or through fixed similarity metrics, so the graph structure does not evolve as the model learns which patient relationships are actually discriminative Lin et al. [2023]. Second, hyperedges connecting groups of patients are typically built once before training as static heuristics Ding et al. [2023], preventing the model from learning the higher-order group relationships that reflect overlapping clinical presentations in neurodegenerative disease.

These limitations make it difficult to reliably distinguish diseases like Alzheimer's and vascular dementia, which share cognitive symptoms but differ substantially in pathology and structural brain signatures Zhou et al. [2023].

Preprint.

Our project addresses this directly. We combine structural MRI features as node-level representations with clinical tabular data from the NACC dataset, including cognitive assessments, vascular risk profiles, and medical history, as dynamically weighted hyperedges. We fuse this hypergraph with a cross-attention transformer so that imaging and clinical modalities can inform each other during classification Yao et al. [2023]. The goal is a model whose disease associations are grounded in both brain structure and clinical profile, making it more generalizable across the overlapping presentations that complicate differential diagnosis. We benchmark against established baselines with a target classification accuracy above 90% Zhang et al. [2024].

This work addresses three questions: whether dynamically learned hyperedge weights improve multi-class differential diagnosis over static population graph approaches; whether fusing structural MRI node features with clinical tabular edge representations improves generalization across overlapping presentations; and how our architecture compares to established graph-based baselines on the NACC Uniform Data Set.

This paper makes three contributions. First, we propose a population hypergraph in which hyperedge weights are learned during training rather than fixed as pre-training heuristics. Second, we introduce a cross-attention fusion mechanism that lets structural MRI features and clinical tabular data jointly inform node representations. Third, we provide a multi-class evaluation on the NACC dataset against GCN, Chebyshev convolutional network, and graph attention network baselines.

The remainder of this paper is organized as follows. The Related Work section surveys graph-based approaches to neurological disease classification, identifying the gaps our work addresses. The Proposed Methodology section then formalizes our approach, detailing the data pipeline, model design, and evaluation plan.

2 Related Work

2.1 Graph Neural Network Foundations

Graph Convolutional Networks (GCNs) established a framework for learning on graph-structured data by aggregating features from a node's local neighborhood through layer-wise transformations Kipf and Welling [2017], forming the basis for medical imaging applications on graph-structured data. For neurodegenerative disorders, recent taxonomies divide these applications into subject-graphs (nodes as brain regions) and population graphs (nodes as patients) Zhou et al. [2023]. Population graphs are particularly effective for multi-subject classification because they allow non-imaging tabular clinical data to be incorporated directly into the edge structure.

2.2 Population Graph Approaches for Disease Classification

How edges are constructed between patients significantly affects classification performance. Connecting patients by clinical scores such as the Clinical Dementia Rating (CDR) yields higher accuracy on Alzheimer's tasks than demographic-variable edges Lin et al. [2023]. Non-imaging features like comorbidities from health records also provide strong diagnostic signals when used to define similarity edges Kazemia et al. [2024], and diffusion maps capture nonlinear manifold structures of high-dimensional MRI data better than standard cosine similarity heuristics Yang et al. [2024].

2.3 Hypergraph Methods in Neurological Disease Diagnosis

Population graphs are limited by pairwise edge construction, which struggles to capture high-order relationships between patient groups. Hypergraph convolutional networks address this by connecting groups of clinically similar patients via hyperedges, improving multi-class disease staging Ding et al. [2023]. Extensions include evolving hypergraph networks with attention mechanisms that weigh neighbors by diagnostic relevance Yang et al. [2022]. More recently, prompted hypergraph neural networks (PHGNNs) use prompt learning to handle missing data and heterogeneous clinical inputs, consistently outperforming static pairwise baselines Liu and Rossi [2025].

2.4 Causal Reasoning and Diagnoses in Neurological Diseases

Many graph-based classifiers rely on static correlations, but there is growing focus on extracting causal relationships to improve clinical explainability and diagnostic robustness. Some frameworks combine spatiotemporal GCNs with causal mediation analyses to validate whether network features reflect true biological markers of progression Zhang et al. [2024]. Others explicitly encode causal directionality between brain regions Cao et al. [2024] or use Granger-causality-inspired networks to disentangle causal and non-causal subgraphs Zheng et al. [2024]. These methods aim to align diagnostic logic with neuropathological biomarkers rather than spurious artifacts.

2.5 Multi-Modal Fusion for Neurological Diagnosis

Fusing 3D structural imaging with tabular clinical data is a central challenge for differential diagnosis. Traditional fusion combines features late in the pipeline, missing early interactions. Modern approaches use dynamic alignments: Dynamic Affine Feature Map Transform (DAFT) rescales and shifts 3D imaging features based on tabular inputs Pölsterl et al. [2021], while hypernetwork approaches let tabular inputs dictate fusion weights Duenias et al. [2024]. Cross-attention mechanisms have proven especially effective by letting tabular features query relevant spatial regions of an MRI scan dynamically. Recent transformer backbones have performed complex differential diagnoses across multiple dementia subtypes, demonstrating that proper multimodal fusion can improve the diagnostic abilities of even human physicians Xue et al. [2024].

3 Methodology

3.1 Problem Formulation

The goal of this project is to perform differential diagnosis of neurodegenerative diseases. Our input represents patients where each patient has two distinct types of data. The first is 3D structural MRI which we will define mathematically as $\mathbf{x}_i^{\text{img}}$. The second is a list of tabular clinical data including cognitive scores and demographic data which we will define as $\mathbf{x}_i^{\text{tab}}$. Our target output of our model is a probability distribution over specific diseases like Alzheimer's or Vascular Dementia. This output diagnosis will be defined as Y.

Our approach assumes that neurodegenerative diseases present with highly overlapping symptom profiles. We assume that analyzing a single MRI on its own is often insufficient for a confident diagnosis, so we formulated this as a population graph problem. Instead of mapping a single patient directly to a diagnosis we map a patient into a hypergraph. In this hypergraph the connections between patients are dynamically defined by similarities in the $\mathbf{x}_i^{\text{tab}}$ clinical profiles. The network then uses the gathered context of the connected population to finalize the diagnostic prediction Y.

We formalize this as follows. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{H})$ be a population hypergraph where $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$ is the set of N patient nodes. Each node v_i is associated with two feature representations:

- $\mathbf{x}_i^{\text{img}} \in \mathbb{R}^d$, a feature vector derived from preprocessing of the patient's 3D structural MRI.
- $\mathbf{x}_i^{\text{tab}} \in \mathbb{R}^k$, a vector of tabular clinical features (cognitive scores, demographics).

Edges and hyperedges in $\mathcal{G}$ are defined as:

- $\mathcal{E} = \{e_{ij} \mid w_{ij} = \text{sim}(\mathbf{x}_i^{\text{tab}}, \mathbf{x}_j^{\text{tab}}) > \tau\}$, pairwise edges weighted by clinical similarity.
- $\mathbf{H} \in \{0, 1\}^{N \times M}$, the hyperedge incidence matrix over M hyperedges, where each hyperedge h_m has a learned weight $w_m \in \mathbb{R}$ updated during training.

Our approach assumes that neurodegenerative diseases present with highly overlapping symptom profiles, and that analyzing a single MRI in isolation is often insufficient for a confident diagnosis. Rather than mapping a single patient directly to a diagnosis, we map each patient into this hypergraph, where connections between patients are dynamically defined by similarities in their $\mathbf{x}^{\text{tab}}$ clinical profiles. The network then uses the gathered context of the connected population to produce the final diagnostic prediction:

$$\hat{y}_i \in \{\text{AD}, \text{VaD}, \text{FTD}, \text{LBD}, \text{MCI}, \text{CN}\} \quad \forall\, v_i \in \mathcal{V} \tag{1}$$

3.2 Data pipeline

Our data is sourced from the National Alzheimer's Coordinating Center (NACC) data repository. For our tabular clinical features $\mathbf{x}_i^{\text{tab}}$ we use the Uniform Data Set (UDS) combined with available neuropathology and genetic data. For our imaging modality $\mathbf{x}_i^{\text{img}}$ we use both the SCAN compliant and mixed protocol 3D structural MRI scans. Patients lacking either a baseline MRI or a core set of clinical assessments are excluded.

Raw MRI scans can be significantly varied depending on the scanner hardware and user decisions. To standardize the $\mathbf{x}_i^{\text{img}}$, we will use the MONAI (Medical Open Network for AI) library. All 3D MRI volumes will be resized to a uniform spatial dimension and we will also perform "skull stripping" to isolate the brain from irrelevant artifacts such as the skull and eyes. We will also apply intensity normalization so the network is not biased by different machine calibrations.

Finally, we will split the data into three subsets: 70% for training, 10% for validation, and 20% for final testing. We will also use a stratified split to address class imbalance. This should guarantee the exact proportional distribution of each specific dementia subtype is maintained across the training, validation, and testing sets.

3.3 Model Architecture

Our proposed architecture has a few distinct moduals that we will combine in 3 distinct ways.

1. **Node Generation**
   An important step is to translate the physical brain scans into a mathematical format. To do this we pass a patient's 3D MRI data ($\mathbf{x}_i^{\text{img}}$) through a visual encoder f_{enc}. This encoder compresses the visual data into a dense embedding.

$$\mathbf{z}_i = f_{\text{enc}}(\mathbf{x}_i^{\text{img}}) \in \mathbb{R}^d \tag{2}$$

   Each patient embedding $\mathbf{z}_i$ acts as a single isolated node in our graph.

2. **Edge Construction**
   In order to combine the tabular clinical data with the MRI images. we need to build our network connections. We feed the tabular clinical data ($\mathbf{x}_i^{\text{tab}}$) into a transformer network. The transformer uses the demographic and cognitive test data to dynamically construct hyperedges. These hyperedges connect groups of patients who share complex similarities. We calculate the weights of these connections using the attention formula:

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right), \quad \mathbf{Q} = T(\mathbf{X}^{\text{tab}})\mathbf{W}_Q, \quad \mathbf{K} = T(\mathbf{X}^{\text{tab}})\mathbf{W}_K \tag{3}$$

   where $T(\cdot)$ denotes the transformer encoding and $\mathbf{W}_Q, \mathbf{W}_K \in \mathbb{R}^{d \times d_k}$ are learned projection matrices.

3. **Hypergraph Convolution**
   Once the nodes are grouped into hyperedges the network can do message passing. This means that information is shared across the connections, updating individual patient's visual embedding based on a weighted summary of the visual features from other similar patients in their hyperedge. The convolution update rule is:

$$\mathbf{Z}^{(l+1)} = \sigma\Big(\mathbf{A}\,\mathbf{Z}^{(1)}\,\mathbf{W}^{(l)}\Big) \tag{4}$$

   where $\sigma(\cdot)$ is a nonlinear activation function and $\mathbf{W}^{(l)}$ is the learnable weight matrix for layer l.

4. **Final Prediction**
After the message passing phase is done, we pass the newly enriched embeddings through a final classification layer. This layer will output the final prediction ($\hat{y}_i$) which is a probability score for each type of dementia.

$$\hat{y}_i = \mathrm{softmax}\Big(\mathbf{W}_{\mathrm{cls}}\,\mathbf{z}_i^{(L)} + \mathbf{b}_{\mathrm{cls}}\Big) \tag{5}$$

where $\mathbf{z}_i^{(L)}$ is the final-layer embedding for patient v_i, and $\mathbf{W}_{\mathrm{cls}}, \mathbf{b}_{\mathrm{cls}}$ are the learned classification parameters.

3.4 Implementation Variants

The core architecture was adapted into three distinct implementations to evaluate different data fusion strategies.

- **AWHGCN (Mongold):** The AWHGCN model follows a two-stage training pipeline that decouples representation learning from graph reasoning. In Stage 1, a 3D DenseNet121 backbone is trained on labeled patients (Groups A and B) using cross-entropy with label smoothing, producing 128-dimensional MRI embeddings via a projection head. In Stage 2, the frozen CNN embeddings are concatenated with 128-dimensional clinical embeddings from a linear tabular projector, yielding 256-dimensional node embeddings per patient. These nodes are organized into a hypergraph structure containing two diagnosis hyperedges (one connecting all AD patients, one connecting all non-AD patients) and N kNN hyperedges (one per patient, linking each to their five nearest neighbors in clinical feature space), with Group C (MCI) nodes participating in kNN edges but excluded from diagnosis edges. The embeddings are then passed through two attention-weighted hypergraph convolution (AWHGConv) layers, where an MLP attention mechanism computes per-node-per-edge weights from the CNN embeddings, ensuring that image-based evidence modulates the influence of graph structure rather than being overridden by it. The model is trained jointly with cross-entropy plus a 0.1-weighted supervised contrastive loss (SupCon) under differential learning rates (CNN at $1\mathrm{e}{-5}$, GNN at $1\mathrm{e}{-3}$), and soft predictions for Group C are recovered transductively as the attention ratio $P(\mathrm{AD} \mid i) = \alpha_{i,\mathrm{AD}}/(\alpha_{i,\mathrm{AD}} + \alpha_{i,\text{non-AD}})$ between a patient's attention to the two diagnosis hyperedges.

- **Hyperfuse (Furlan):** The HyperFuse model trains end to end through 4 distinct stages. First is the *CNN layer* where the model passes the image through a *3D Convolution layer*, a *BatchNorm3D layer*, and a *ReLU activation function* to extract **spatial features** and **raw patches**. The **raw patches** are then passed through the *hypergraph network* based on an **adjacency matrix** pre-computed from the clinical features of every patient. Finally, the resulting **blended features** are used to modulate the **spatial features** from the original *CNN layer*, and a *residual connection* is added as well. The final features are then either passed to the next layer, or to the classifier which outputs the models final predictions.

- **[GCMHN](Noone):** The Gated Cross-Modal Hypergraph Network (GCMHN) is trained through 5 stages. Firstly, the 3D MRI scan is run through a simple 3D CNN which encodes it down into a vector representation (Z). Then, in parallel, the clinical tabular data is used to construct soft hyper-edge memberships in H_{clin}. The MRI embeddings are used to then create a gate (G) which will aid in deciding which hyper-edges to suppress and which to enhance $G \in [0, 1]$. Then, once the MRI gate is fused with the clinical tabular data, this new H is passed into the soft hypergraph convolution stage which refines the Z embeddings by passing information between patients which are similar. This stage is similar to attention conceptually, since information is aggregated for each patient from similar patients. However, the similarity scores are not computed pairwise, but are derived from the shared hyper-edge memberships defined by clinical data, and then refined by the MRI features. Finally, the refined embeddings are transformed into logits which are used for the multi-class disease prediction.

4 Experiments

4.1 Experimental setup

All models are evaluated on different 200-300 patient NACC cohorts consisting of six categories equally distributed across Normal Cognition, Mild Cognitive Impairment, Alzheimer's disease, Vascular Dementia, Lewy Body Dementia, and Frontotemporal Dementia. 70% of the data was then partitioned for training, leaving 10% for validation, and 20% for testing.

We measure our model's success using two common metrics: overall correctness (accuracy) and a balanced score that treats all categories equally (macro-F1).

To evaluate how the models handle overlapping symptoms between closely related diseases like Alzheimer's and Mild Cognitive Impairment, we analyzed the model's prediction confidence (outputting a percentage chance rather than a strict yes/no). We then organized these soft probability scores based on the actual underlying medical causes. We hid these underlying causes from the model until *after* the experiment to analyze our final results. We used them to teach or train the AI.

Models train on a single graphics card (10-16,GB VRAM). Training times vary significantly by architecture: each AWHGCN fold takes approximately 5 minutes, while the end-to-end HyperFuseNet pipeline takes approximately 35 minutes to complete 20 epochs.

We compare our architectures against several baselines: Logistic regression and XGBoost on the 23 tabular features, plus a CNN-only baseline (Stage 1 DenseNet121) and a CNN+MLP late-fusion baseline.

4.2 Compared models

We benchmark three architectures developed independently within the project, each addressing differential dementia diagnosis from the same NACC cohort:

- **AWHGCN (Mongold):** Hypergraph GNN with CNN-parameterized hyperedge attention and end-to-end multimodal fusion, positioned against prior static-hyperedge dementia work Ding et al. [2023], Yang et al. [2022], Liu and Rossi [2025].
- **HyperFuseNet (Furlan):** An inductive hypergraph architecture trained end-to-end that learns general rules about how to build connections between nodes unseen during training.
- **[GCMHN] (Noone)**. A Gated cross-modal hypergraph network which uses clinical tabular data to construct soft patient relationships, and MRI-derived gating to refine them. Which enables attention-like message passing over multi-modal patient graphs.

All three models are evaluated on identical CV splits to enable direct comparison.

4.3 Main results

Table 1: Model performance on the 20 percent holdout test set.

Model	Accuracy	Macro-F1
LR-tabular	0.595	0.591
XGB-tabular	0.650	0.624
CNN-only	0.670	0.659
CNN+MLP	0.660	0.648
AWHGCN (Mongold)	0.598	0.591
Hyperfuse (Furlan)	0.348	0.312
GCMHN (Noone)	0.20	0.13

AWHGCN findings: The AWHGCN model achieved a mean macro-F1 score of 0.59. This means it only performed as well as our simplest tabular baseline, and actually underperformed compared to both the CNN-only (0.66) and XGBoost (0.62) models. An analysis of the model's errors reveals a "class collapse" issue, where the model overwhelmingly defaulted to predicting Alzheimer's Disease (AD). While it successfully identified 76% of true AD cases, it incorrectly predicted AD for 54% to

82% of patients who actually had other conditions. Its confidence scores were unhelpful, hovering around a 0.5 coin-toss (within $\pm 3\%$) regardless of the patient's actual underlying condition, including those with Mild Cognitive Impairment (MCI).

For the 100 patients in Group $\mathcal{C}$, the model's predicted probability of AD ($\hat{p}_{\mathrm{AD}}$) was confined to a narrow margin between 0.44 and 0.49, indicating a complete inability to distinguish between individual cases. While previous research highlights the effectiveness of hypergraph methods on much larger cohorts, our findings suggest that a dataset of $n = 300$ falls short of the required sample size. Without sufficient data, the model's attention mechanism fails to learn the complex patterns necessary to make distinct, reliable predictions.

Hyperfuse findings: The Hyperfuse model achieved a Macro-F1 score of 0.312 and an accuracy of 34.8%. This is significantly worse than every baseline model, but outperformed a random guessing strategy that we would expect to achieve around 16.67% accuracy. A large limitation of the Hyperfuse model is that the complex multi-layer data fusion steps that are trained end-to-end require a lot of data to train. Because of external factors like limited hardware and time, it was infeasible to train the Hyperfuse model on as much data as would be necessary to truly test its limitations.

GCMHN findings. On the 20% held out test set, the GCMHN model resulted in an accuracy of 20% and a macro-f1 of $\approx 0.13\%$, meaning that the model was essentially just distributing probability relatively evenly over all classes when making predictions. Now, this lack of performance, is likely due to the lack of data magnitude in processes along with the fact that the model utilized batching.

5 Discussion

5.1 Cross-model analysis

Across all three architectures, the dominant failure mode was a collapse toward the majority or most prototypical class rather than discriminative use of learned graph structure. AWHG-CN converged to the uniform $\hat{p}_{\mathrm{AD}} \approx 0.46$ across group $\mathcal{C}$. HyperFuseNet's accuracy of 34.8% sat above the floor of class collapse and GCMHN's 20% accuracy is essentially indistinguishable from uniform random guessing across six classes. The fact that all three independently designed architectures (parameterized HyperEdges attention, end-to-end inductive fusion, and gate cross-modal message passing) converged on the same qualitative failure pattern strongly suggests that the bottleneck is not architectural but data-driven. A two to three-hundred patient cohort is insufficient to learn the high-order interpatient relationships that the hypergraph formulation is designed to exploit. The simpler CNN-only baseline (0.66 macro-F1) and XGBoost on Tabular Features (0.62) outperformed all three hypergraph variants indicating that at this sample size, the additional inductive capacity of the graph layers acts as a regularization burden rather than a representational asset.

5.2 The challenge of End-to-End Multi-modal Fusion

Combining 3D brain scans and clinical data in a single model is incredibly difficult. The main challenge is balancing two completely different types of information. We have massive 3D MRI images that take up a lot of memory and simple clinical numbers like age and blood pressure. Since these features are so different in terms of both information density and signal intensity we cannot compare these features directly through the same mechanisms. This is why we need to train a complex multi-modal system to handle both information types separately and blend their embedding representations intelligently.

Additionally, extracting useful patterns directly from 3D pixels requires a massive amount of computer memory. Because the model also tries to build a dynamic graph comparing patients using their clinical data, the memory limits are hit almost immediately. This forces us to find work arounds like projecting the patches into smaller embeddings or using very small batch sizes.

In the Hyperfuse model in particular, the spatial gating mechanism tries to highlight specific areas of the brain based on what it learns from other patients in the batch. However because the batch size is so small the model never sees enough examples at one time to learn stable rules. It definitely learns real patterns since it beats random guessing but the architecture is simply too heavy to reach its full potential on our limited hardware and dataset.

5.3 Computational Bottlenecks and Sample Inefficiency

Comparing our models to recent studies that have used similarly complex AI systems on similar patient groups gives us a hint as to why our models failed to exceed the baselines. Namely, those studies used substantially larger datasets to train their AI, which allowed them to achieve very high performance scores between 0.85 and 0.95. This demonstrates that our models could potentially achieve much better results if we were able to train them on similarly large datasets. However, We are not trying to compete with those massive data heavy approaches in this specific study, but rather demonstrate the effectiveness of our specific inductive architectures on a strictly filtered and highly pure dataset of 200 to 300 patients.

5.4 Future Directions

Several future research directions follow directly from the failure modes identified above. First, scaling the cohort beyond the current 200-300 patient subset is essential. This would test whether the end-to-end hypergraph architectures recover the performance trajectory reported in prior work on larger datasets Ding et al. [2023], Liu and Rossi [2025]. Specifically, a larger patient population would provide the statistical power necessary to stabilize the dynamic spatial gating mechanisms, potentially resolving the class-collapse observed in under-represented categories. Second, to support this scale without exceeding VRAM hardware limitations, future iterations should explore decoupled representation learning. By pre-training and freezing the 3D convolutional feature extractor before constructing the clinical adjacency matrix, the network could bypass restrictive batch limits. This would allow the hypergraph to perform message passing across the entire global patient population simultaneously, merging the inductive strengths of dynamic graph building with the computational efficiency required for clinical deployment.

6 Conclusion

6.1 Summary of Primary Contributions

This work introduced and evaluated three independently designed hypergraph architectures for differential diagnosis of neuro-degenerative disease on the NACC Uniform Data Set: AWHGCN, which uses CNN-parameterized attention to weight hyperedges between diagnosis-defined and kNN-defined patient groups; HyperFuseNet, an end-to-end inductive architecture that fuses 3D convolutional features with hypergraph-blended representations through residual modulation; and GCMHN, a gated cross-modal network that uses MRI-derived gates to refine clinical-tabular hyperedge memberships before message passing. All three were evaluated on identical 200–300 patient cohorts spanning six diagnostic categories, with comparison against logistic regression, XGBoost, CNN-only, and CNN+MLP late-fusion baselines. Our central methodological contribution is the framing of hyperedge weights as learnable, attention-modulated quantities rather than fixed pre-training heuristics, evaluated across three distinct fusion strategies on the same cohort.

6.2 Implications for Clinical AI

Our results are cautionary on two to three hundred patients Simpler tabular and CNN baselines outperformed all three hypergraph variants. Architectural complexity should be matched to cohort scale. The class collapse pattern, where the models default to AD with a near coin flip confidence, is clinically dangerous given that vascular, Lewy body, and frontotemporal dementias have meaningfully different treatment paths. Calibration, not just top one accuracy, must be a first-class evaluation criterion.

6.3 Limitations and Future Work

Three limitations bound our conclusions: cohort size (30–50 per class is below the regime where prior hypergraph work succeeds), hardware constraints (single-GPU training forced small batches and low-dimensional embeddings), and strict purity filters that may have removed the ambiguous cases where multi-modal hypergraphs help most. Future work should scale to multi-center NACC cohorts on multi-GPU hardware with relaxed inclusion criteria.

Author Contributions

J.M. Worked on Introduction, Methodology, Discussion, and Conclusion Sections.

A.F. Worked on the Methodology, and Discussion Sections as well as editing on the Conclusion and Experiments sections.

D.N. Worked on Related Works section.

References

J. Cao, L. Yang, P. G. Sarrigiannis, D. Blackburn, and Y. Zhao. Dementia classification using a graph neural network on imaging of effective brain connectivity. *Computers in Biology and Medicine*, 2024. URL https://www.sciencedirect.com/science/article/pii/S0010482523011666?

Yi Ding, Hao Zhang, Tao Wang, Zhiyu Li, Ling Ma, Yifan Tao, and Dinggang Shen. Hypergraph convolutional network for longitudinal data analysis in Alzheimer's disease. *Computers in Biology and Medicine*, 167:107626, 2023. doi: 10.1016/j.compbiomed.2023.107626.

Daniel Duenias et al. HyperFusion: A hypernetwork approach to multimodal integration of tabular and medical imaging data for predictive modeling. *arXiv preprint arXiv:2403.13319*, 2024. URL https://arxiv.org/abs/2403.13319.

Maryam Kazemia, Yijun Zhao, Fei Hua, and Xiaobo Zhang. Comorbidity-based framework for Alzheimer's disease classification using graph neural networks. *Scientific Reports*, 14:21185, 2024. doi: 10.1038/s41598-024-72321-2.

Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations (ICLR)*, 2017. URL https://arxiv.org/abs/1609.02907.

Lan Lin, Minjie Xiong, Ge Zhang, Wenjie Kang, Shen Sun, and Shuicai Wu. A convolutional neural network and graph convolutional network based framework for AD classification. *Sensors*, 23(4): 1914, 2023. doi: 10.3390/s23041914.

Chengyi Liu and Luca Rossi. PHGNN: A novel prompted hypergraph neural network to diagnose Alzheimer's disease. *arXiv preprint arXiv:2503.14577*, 2025. URL https://arxiv.org/abs/2503.14577.

Sebastian Pölsterl, Thomas N. Wolf, and Christian Wachinger. Combining 3D image and tabular data via the dynamic affine feature map transform. *arXiv preprint arXiv:2107.05990*, 2021. URL https://arxiv.org/abs/2107.05990.

Chonghua Xue, Sahana S. Kowshik, Diala Lteif, et al. AI-based differential diagnosis of dementia etiologies on multimodal data. *Nature Medicine*, 30:2977–2989, 2024. URL https://www.nature.com/articles/s41591-024-03118-z.

Changqing Yang, Paul Wein, Melissa Holm, Anna Thomas, and Islem Rekik. An evolving hypergraph convolutional network for the diagnosis of Alzheimer's disease. *Frontiers in Aging Neuroscience*, 14:1004927, 2022. doi: 10.3389/fnagi.2022.1004927.

Zhi Yang, Kang Li, Haitao Gan, Zhongwei Huang, Ming Shi, and Ran Zhou. An Alzheimer's disease classification network based on MRI utilizing diffusion maps for multi-scale feature fusion in graph convolution. *Mathematical Biosciences and Engineering*, 21(1):1554–1572, 2024. doi: 10.3934/mbe.2024067.

Dongren Yao, Jing Sui, En Yang, Pew-Thian Yap, Dinggang Shen, and Mingxia Liu. Multi-modal graph neural network for early diagnosis of Alzheimer's disease from sMRI and PET scans. *Computers in Biology and Medicine*, 164:107310, 2023. doi: 10.1016/j.compbiomed.2023.107310.

Peng Zhang, Haotian Lin, Yingqi Guo, Meng Liao, Rongbo Luo, Zhijun Qiao, Zhiyang Liu, and Daoqiang Zhang. A novel spatiotemporal GCN framework for functional connectivity biomarker identification of Alzheimer's disease. *Alzheimer's Research & Therapy*, 16:57, 2024. doi: 10.1186/s13195-024-01425-8.

K. Zheng, S. Yu, and B. Chen. CI-GNN: A Granger causality-inspired graph neural network for interpretable brain network-based psychiatric diagnosis. *Neural Networks*, 2024. URL https://www.sciencedirect.com/science/article/abs/pii/S0893608024000716?

Zheyuan Zhou, Wenxiang Li, Xiaoting Sheng, Shuying Zheng, Yang Zhou, Jian Liu, Xin Li, and Zhongxun Sun. The combination of a graph neural network technique and brain imaging to diagnose neurological disorders: A review and outlook. *Biomedicines*, 11(10):2832, 2023. doi: 10.3390/biomedicines11102832.